Cyber Environment

One word, six senses, counted twice

What is a cyber environment? The word does six separate jobs across the two places I work from most, and neither place ever says which job is meant. This file counts them.

Fifty-six occurrences of the token, forty-nine of them carried into the census, six senses, and twenty sentences where a second pass through the same words lands on a different sense than the first. The twenty stay ambiguous. They are the entries worth having.

Contents

1 The two corpora

Corpus A is the eleven pages of Cyber Environments & Benchmarks on the site. Seven of the eleven carry prose: the README and the pages for HackIT, CybORG, Daedalus, Cyber Wheel, Cyber Battle Field, and CyberVAN. Four carry no prose. CyGym and FireWheel are bare headings, the LLM-based Cyber Scenario Generator page is a heading and an arXiv link, and MDP Profiling of Cyber Environments has a front matter that reads *Stay Tuned...* and no body. Twenty-three tokens.

Corpus B is the realism evaluation, fifteen environments scored on eleven dimensions with a written justification in every cell, three hundred and ninety-three lines. Thirty-three tokens.

Seven tokens are set aside before the census starts. Two are file paths in corpus A and one is a CSV filename in corpus B. The line *Work of mine that runs on this environment* appears identically on five pages of corpus A and is folded to a single row. Forty-nine rows remain.

2 Six senses

Each entry gives the definition, the nearest sense it is confused with, a yes-or-no test that decides between them, and where it lands in the corpora.

1. The package

Definition. A named, distributable piece of software that you clone, configure, and run, with a version and a repository.

Distinguishes it from sense 2. The package exists before anyone decides who is playing. Cyberwheel is one package and supports red-only training, blue-only training, and simultaneous training of both.

Operational test. Can you name its version without naming an agent?

Where it appears. The dominant sense, twenty-nine rows of forty-nine, and the default reading of every heading in both corpora.

> *"Cyberwheel is a reinforcement learning simulation environment for training and evaluating autonomous cyber defence models, built for modularity."*
>
> blog, cyber-wheel.md

2. The decision problem

Definition. Everything outside the learner: observation space, action space, reward, transition dynamics, episode boundary. The environment in the sense reinforcement learning means it.

Distinguishes it from sense 1. One package presents as many decision problems as it has agent seats and opponent configurations.

Operational test. Does swapping the opponent, with no change to the code, give you a different one? If yes, this is the sense in play. CybORG with the session-removing defender and CybORG with the host-restoring defender are two decision problems and one package.

> *"The restoring defender is what makes the environment non-stationary from the attacker's side."*

blog, cyborg.md

> *"the reward function, observation space, and action space can be redefined without rewriting the environment"*
>
> blog, cyber-wheel.md

That second sentence is the clearest place the two senses stand side by side. Three sense-2 objects are named, and the noun at the end of the same clause is sense 1.

3. The modeled system

Definition. The estate being represented. Hosts, subnets, services, accounts, directories, logs, the traffic a workday produces. What the eleven realism dimensions score.

Distinguishes it from sense 1. Realism is a property of this sense and never of the package. A well-engineered package can model almost nothing.

Operational test. Can you ask whether a real technique would work in it, Kerberoasting say, and get an answer that is not about the code?

> *"Multi-forest AD with trust relationships. Kerberoastable accounts, ADCS certificate abuse, GPO misconfigurations, tiered administration (partially). The most identity-rich environment in the catalog."*
>
> realism evaluation, GOAD, Identity

4. The substrate

Definition. The provisioned infrastructure underneath. Virtual machines, KVM or Vagrant or Terraform, the range, the wire. What is there before any scenario is loaded onto it.

Distinguishes it from sense 3. The substrate is what the scenario runs on; the modeled system is what the scenario says. CALDERA has no sense-3 environment of its own and inherits one from whatever it is deployed against.

Operational test. Would it still be there if you deleted every scenario file?

> *"CALDERA is a tool that operates on real infrastructure, not a self-contained environment"*
>
> realism evaluation, CALDERA, profile

> *"it inherits OS / service realism from the deployment infrastructure"*
>
> realism evaluation, CALDERA, profile

5. The experimental role

Definition. A position one environment occupies relative to another inside a particular study. Source, target, far target, reference point.

Distinguishes it from sense 1. This is a property of the experiment and not of the software. Cyber Wheel is a good source because of the pairing, not because of anything in its repository.

Operational test. Does the word carry a modifier that names a position in a study rather than a property of a system?

> *"For transfer work it makes a good source environment"*

blog, cyber-wheel.md

"its action space only partially aligns with a kill-chain source environment and its observation space is substantially different, which makes it the larger of the two domain gaps tested"

blog, cyber-battle-field.md

"A high-fidelity reference point rather than a training environment for this work."

blog, cybervan.md

6. The third party

Definition. What is neither the attacker nor the defender. Benign users, service load, maintenance windows, background traffic. The green agent.

Distinguishes it from sense 2, and this is the sharpest collision in the file. Under sense 2 the opponent is inside the environment by definition. Under sense 6 the opponent is explicitly outside it and the environment is the third term in a list of three.

Operational test. If you removed it, would the attacker be the only source of activity?

"No simulated users or background traffic. The attacker agent is the only source of activity."

realism evaluation, CyberBattleSim, Benign Activity

"Green agents simulate benign activity."

realism evaluation, Cyberwheel, Benign Activity

The same cell adds that most simulators have no benign activity at all, and the summary table agrees. Of the fifteen environments, two score partial on Benign Activity, twelve score absent, and none scores full.

"This makes it uniquely suited for studying attacker-defender-environment interaction at the simulation level."

realism evaluation, key observations

3 The census

Forty-nine occurrences. Column I is the first assignment. Column II is a second pass made from the sentences alone, without column I in view. Twenty rows disagree and are marked.

#	Locator	The words	I	II
Corpus A, the eleven blog pages				
1	README, title	Cyber Environments & Benchmarks	1	1
2	README	a claim about the environment it was measured in	1	2*
3	README, figure	compared on observation richness, action space size, and fidelity	2	3*
4	README, caption	Three axes, six environments. No single ordering.	1	1
5	README	The environments closest to a real system are the ones you can least afford to train in	3	1*
6	cyber-battle-field	a kill-chain source environment	5	2*

#	Locator	The words	I	II
7	cyber-wheel	A configurable RL environment where the defender's main move is deception	1	1
8	cyber-wheel, figure	the environment assembled from configuration	3	1*
9	cyber-wheel	a reinforcement learning simulation environment for training and evaluating autonomous cyber defence models	1	1
10	cyber-wheel	redefined without rewriting the environment	1	1
11	cyber-wheel	it makes a good source environment	5	1*
12	cybervan	rather than a training environment for this work	5	5
13	cybervan, link	When We Say A Realistic Cyber Environment	3	3
14	cyborg	makes the environment non-stationary from the attacker's side	2	2
15	daedalus	The tool that runs this environment	4	1*
16	five pages	Work of mine that runs on this environment	1	1
17	mdp-profiling, title	MDP Profiling of Cyber Environments	1	2*
Corpus B, the realism evaluation				
18	title	Full Realism Evaluation: 15 Environments by 11 Dimensions	1	1
19	heading	Per-Environment Evaluations	1	1
20	FARLAND, source	Network Environment Design for Autonomous Cyberdefense	1	3*
21	CyberShield, source	A Competitive Simulation Environment for Training AI in Cybersecurity	1	1
22	CYE, heading	CYE, Cyber Environment	1	1
23	CYE, source	Cyber Environment Test Framework for Simulating Command and Control Attack Methods	1	1
24	CYE, Topological	multi-node configurations simulating cloud environments	3	3
25	CYE, External	Isolated network environment. No external DNS, OSINT, or C2.	3	3
26	CyGIL, source	Unified Emulation-Simulation Training Environment for Autonomous Cyber Agents	1	1
27	Cyberwheel, source	Towards a High Fidelity Training Environment for Autonomous Cyber Defense Agents	1	1
28	PenGym, Telemetry	No structured log collection, SIEM, or EDR documented as part of the environment	1	3*
29	PenGym, profile	Highest action/observation realism among RL-oriented environments	1	1
30	CALDERA, Observation	This is by design (adversary emulation tool, not RL environment)	2	1*
31	CALDERA, profile	Highest action realism of any environment in the catalog	1	1
32	CALDERA, profile	a tool that operates on real infrastructure, not a self-contained environment	1	4*
33	GOAD, Identity	The most identity-rich environment in the catalog	3	3
34	GOAD, Defensive	The environment is a target, not a defended target	3	6*
35	GOAD, Action	fully autonomous LLM-driven penetration of the environment	3	4*
36	GOAD, profile	purely a target environment	5	3*
37	SCORPION, OS	Virtualized environments imply real OS	4	4
38	SCORPION, Action	Real virtualized environments imply real tool execution	4	4
39	HTB/CTF, profile	Useful as action-realism benchmarks, not as environment-realism benchmarks	3	1*
40	aggregate table	column head, Environment	1	1
41	key observations	No environment achieves full on more than 5 dimensions	1	1
42	key observations	No environment scores full on Topological, Temporal Dynamics, Defensive, Benign Activity, Telemetry Fidelity, or External Ecosystem	1	3*
43	key observations	The emulation/real-infrastructure environments split between real-software realism and defensive realism	1	1

#	Locator	The words	I	II
44	key observations	The environments where agents can act realistically are the ones where defenders cannot operate realistically	1	3*
45	key observations	Cyberwheel is the only environment with documented benign-activity modeling	1	1
46	key observations	studying attacker-defender-environment interaction at the simulation level	6	6
47	key observations	No environment scores full.	1	1
48	key observations, Vyas	training environments systematically lack conditions for defender evaluation	1	6*
49	key observations	No environment models external DNS, OSINT, C2 infrastructure, or supply chain	1	3*

Sense counts on the first pass: package 29, modeled system 9, role 4, substrate 3, decision problem 3, third party 1. The package sense takes more than half the corpus on its own, and the third party sense is carried by a single sentence in three hundred and ninety-three lines.

4 Where the two passes disagree

Twenty of forty-nine. Six are set out below. The rest follow the same shapes.

> **The sentence the whole page turns on**
>
> *"Every claim about an autonomous cyber agent is a claim about the environment it was measured in."* README of corpus A, first line of body text.
>
> Read as sense 1 it says results do not travel between packages, which is a reproducibility claim and mild. Read as sense 2 it says a result does not survive a change of opponent inside the same package, which is a much stronger claim and the one the CybORG page actually supports two clicks later. The sentence is the thesis of the whole tree and it does not say which.

> **Fidelity against tractability, on one noun**
>
> *"Notice that fidelity and tractability pull against each other. The environments closest to a real system are the ones you can least afford to train in."* README of corpus A.
>
> Closeness to a real system is a property of sense 3. Affordability of training is a property of sense 2. The noun in the middle is doing both jobs in one sentence, which is why the trade sounds like a law rather than a fact about current engineering.

> **Whether CALDERA is an environment at all**
>
> *"This is by design (adversary emulation tool, not RL environment)."* Corpus B, CALDERA, Observation dimension.
>
> Under sense 2, CALDERA is disqualified because it has no observation model, and the justification in that very cell gives exactly that reason: centralized complete reporting, no partial observability, no noise. Under sense 1 it is disqualified because it ships no scenario, and its profile line gives that reason instead: it is *"a tool that operates on real infrastructure, not a self-contained environment"*. Two disqualifications, two senses, one row of the catalog, and CALDERA is scored on all eleven dimensions regardless.

A target with no defender in it

"The environment is a target, not a defended target." Corpus B, GOAD, Defensive dimension. Sense 3 reading: the modeled estate carries no SIEM, no EDR, no detection. Sense 6 reading: there is nobody in there but the attacker. GOAD's cells support both, since Defensive and Benign Activity and Telemetry are all absent. The two are not the same absence. One is a missing subsystem and the other is a missing occupant, and a fix for one is not a fix for the other.

The benchmark sentence that uses the word twice at two scales

"Useful as action-realism benchmarks, not as environment-realism benchmarks." Corpus B, HTB/CTF profile.
Here *environment-realism* is the narrow sense 3, the estate, set against action realism as one dimension among eleven. But the document that sentence closes is titled *15 Environments by 11 Dimensions*, where the same word is sense 1 and the whole of which action realism is a part. The word names the whole in the title and a part of the whole in the conclusion.

Which environments the defender is missing from

"training environments systematically lack conditions for defender evaluation." Vyas et al. 2025, quoted in corpus B.
Sense 1: the packages do not ship the conditions. Sense 6: the modeled workplaces have no ordinary occupants, so there is no baseline to alert against and nothing for a false positive to be false about. The second is a claim about what is being represented and the first is a claim about what has been built. Cyberwheel's green agents answer the second and not the first.

Two patterns run through the twenty. Fifteen of them involve sense 1 on one side, which says the package name is the reading that swallows the others: when a sentence is doing something more specific, the specific work is in the other column. Eight of the twenty are the same pair, package against modeled system, and six of those eight are sentences that attribute realism or its absence to a repository.

The other pattern is what the passes agreed on. Both passes landed in the same place on every occurrence where the sentence names who is on the other side of the boundary. *From the attacker's side*, in the CybORG entry. *Attacker-defender-environment*, in the key observations. *The attacker agent is the only source of activity*, in the CyberBattleSim cell. Naming the second party is what fixes the word, and it costs four words.

The twenty are recorded as ambiguous and left that way. A sentence that supports two senses is evidence about the corpus, not a defect in the corpus.

5 The neighbor words

Both corpora reach for a second word constantly, and the second word is not the same word in the two of them.

	Corpus A	Corpus B
environment	23	33
fidelity	5	38
realism	0	16
simulation	3	16
simulator	4	11
emulation	1	17
scenario	2	13
target	3	11
range	1	5
gym	1	2
benchmark	1	2
substrate	2	0
platform	0	4

5.1 Fidelity and realism are the same word in two dialects

Corpus A says fidelity five times and never says realism. Corpus B says both, and splits them. Thirty-five of the thirty-eight fidelity tokens in corpus B are inside the dimension names *Service Fidelity*, *OS Fidelity*, and *Telemetry Fidelity*, where fidelity is a property of one named layer. Realism is what the whole gets: defensive realism, identity realism, action realism, temporal realism, and the title of the document.

So the two corpora have the same distinction and reversed labels. Corpus A uses fidelity for the aggregate that corpus B calls realism, in *"fidelity and tractability pull against each other"* and *"CyberVAN aims at the highest fidelity short of deploying the real thing"*. Anyone reading the two together will read the CyberVAN sentence as a claim about eleven dimensions at once, and it is not one.

5.2 Scenario carries two senses of its own

In corpus A, *"The CAGE Challenge 2 scenario is the version most of this work uses"* makes a scenario a configuration of one package: thirteen hosts, three subnets, a reward table. In corpus B, the Admin@338 scenario in the CYE entry is an attack script, a sequence of adversary behaviour with phishing in it. Corpus B uses both, sometimes in one entry.

5.3 Substrate and platform never meet

Corpus A's word for what a challenge is built on is substrate, twice, both times about CybORG under CAGE. Corpus B never says it, and says platform four times, three of them about the human-facing products, HTB and SCORPION, and the fourth about an operating system. The two vocabularies are not in contact and describe adjacent things.

6 Three communities, three defaults

The question this entry answers is what to do when the same word arrives from three directions. The corpora contain all three, and each community's default sense comes with its own idea of what would make an environment more realistic. The three answers do not overlap at all.

Reinforcement learning defaults to sense 2. The environment is whatever returns the next observation and reward. Realism means the decision problem is hard in the right way: partial observability, non-stationarity, delayed credit. The corpus B phrase for a package that fails this is blunt, *"adversary emulation tool, not RL environment"*, and the failing property is a complete report rather than a poor estate.

Security operations defaults to sense 3, with sense 4 close behind. The environment is the estate. Realism means a real technique would work, and the eleven dimensions are the operational form of that: topology, service, OS, identity, temporal, defensive, benign activity, telemetry, action, observation, external. Nothing in that list is about an interface.

Human factors defaults to the task setting a person is put in. HackIT is the corpus's clearest case. *"HackIT exists to put real people in front of a network and watch what they do when some of it is fake."* What it buys is named in the next paragraph and it is not a dimension score: *"an attacker who is actually surprised."* Its constraint is stated as a warning rather than as a missing capability, *"the loop runs at human speed"*.

The SCORPION entry is where the third default collides with the second in writing. The range is scored unknown on ten of eleven dimensions, and its one documented instrumentation stream is thrown out with a sentence that is doing lexicography: *"this is learner telemetry, not security telemetry within the range"*. Same word, two populations being measured, and the evaluation has to say so out loud to keep the column honest.

The diagnostic that separates the three is one question. Ask what would make this environment more realistic. The reinforcement learning answer adds noise and a moving opponent. The operations answer adds Active Directory, event forwarding, and people doing their jobs. The human factors answer adds task pressure and a cover story the participant believes. A conversation in which those three answers are all called realism will not converge, and none of the three is wrong.

7 The five dimensions

The dictionary page places terms as regions on five axes: Object, Epistemic operation, Specificity, Purpose, Fidelity. Environment is already a coordinate value on the first of them, the far end of *actor, behavior, attack possibilities, environment*, and is also the headword here. That is worth knowing before the placement is read: the axis was built to place adversary terms, and this term sits at its endpoint.

Sense	Object	Epistemic operation	Fidelity
1. Package	environment	simulate / enact	spans the axis
2. Decision problem	environment	predict	abstract
3. Modeled system	attack possibilities	describe	the axis itself
4. Substrate	environment	enact	executable
5. Role	does not place	none	none
6. Third party	behavior	simulate	abstract

Two of the five axes do not survive the transfer, and both failures are informative.

Specificity does not carry over. Its scale is written in adversary vocabulary, *generic adversary, adversary class, particular actor*, and none of the six senses is an adversary. The scale that would do the same work here is scenario specificity: a generic enterprise, a sector, a named organization. That is a different axis with the same shape, and putting both on one page will need it said.

Purpose does not separate the senses. All six are used for defensive evaluation across both corpora. It separates the environments from each other, which is a different job. GOAD has no defensive stack and is used for attacker evaluation; HackIT is used for behavioral study. Purpose is an axis for the instances, not for the senses.

Fidelity fails the convexity test on senses 1 and 4. Sense 1 and sense 4 sit at opposite ends of the abstract-to-executable axis, and NASimEmu sits between them: one shared interface with a simulation mode and an emulation mode, scored partial on service and OS fidelity in corpus B. The midpoint is an instance of both senses rather than of neither, so the axis is not what separates them. What separates them is whether the thing ships a scenario. Fidelity is doing real work elsewhere, sorting the fifteen environments into the three categories, and it should not be asked to do this as well.

8 What to say instead

Six senses, six phrases, none longer than the word it replaces.

- Sense 1, name the package and the version. *CybORG at CAGE Challenge 2.*
- Sense 2, name the seat and the opponent. *The blue seat against B-line.* If the opponent changes, say so, because the decision problem changed.
- Sense 3, say the estate, or say the network, and score it on the eleven dimensions rather than on one word.
- Sense 4, say the substrate, or say the infrastructure. Corpus A already does.
- Sense 5, say source and target, and say which study. The role belongs to the pairing.
- Sense 6, say benign activity, or say green agents, and never let it be carried by the bare noun. This is the sense the bare noun destroys most completely, because sense 2 puts the opponent inside the same word.

The page that would settle sense 2 is already on the site and has no body yet. It is called MDP Profiling of Cyber Environments, and the fidelity question belongs on the one next door, When We Say A Realistic Cyber Environment.